Lesson notes · for you

Find and fix bugs

Programming · Level 1 · Lesson 4

Time	Materials
20 min	Paper and a pencil; Arrow cards; A phone (optional)

Learning goals

- Say what a bug is.
- Find the first step that goes wrong.
- Read a robot program written in words.

Activity: Bug detectives

1. Warm-up (3 min): Your teacher, or a friend, acts as a robot and follows a program with one bug. Shout “Stop!” at the first wrong move.
2. Write (7 min): Draw a small grid and write a program for it, with cards or words like forward(2). Hide one bug in it.
3. Hunt (7 min): Swap with a partner. Move a coin along their program, one step at a time. Stop at the first wrong step and fix it.
4. Share (3 min): Tell the class how you found the bug. Did you change only one thing?

A coin or a bottle top makes a good robot on a paper grid.

Discussion

- Why start with the first wrong step?
- How did it feel to find a bug?
- Why is it OK to make bugs?

Check yourself

- Write the four detective steps.

Try it this week

Next time something goes wrong in your homework, be a detective: find the first wrong step, change one thing, and check again.

Worksheet

Find and fix bugs

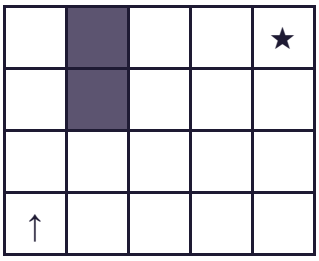
Programming · Level 1 · Lesson 4

Name _____ Class _____ Date _____

1 **Your robot bumps into a wall. What do you do first?**

◯ Add more cards ◯ Throw all the cards away ◯ Find the first wrong card

2 **Sami's program is: forward × 3, turn right, forward × 4. His robot bumps into the wall. Build a program that works, with 5 cards at most.**



Cut out cards (↑ forward, ↶ turn left, ↷ turn right) and lay out a program that takes the robot to the ★. Write it here:

3 **The robot must go forward 2 squares, turn right, go forward 3, turn left, and go forward 1. Put the program in order. One line doesn't belong.**

Number the lines in order (1, 2, 3…). Cross out any line that doesn't belong.

☐ `forward(1)`

☐ `forward(3)`

☐ `forward(2)`

☐ `right()`

☐ `forward(4)`

☐ `left()`

4 **The robot should go forward 2, turn left, then go forward 1. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  forward(2)
2  right()
3  forward(1)
```

5 **Good programmers never make bugs.**

◯ True ◯ False

✎ **Think of a time something went wrong for you: a drawing, a game or a sum. How did you find the mistake? What would the detective do?**

Check your answers

Find and fix bugs

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1 **Find the first wrong card**

Find the first wrong card, change it, and test again.

2 **Any program that takes the robot to the ★ with 5 cards or fewer.**

Sami's robot goes up too far, then turns into the wall.

3 **Top to bottom: 5, 3, 1, 2, X, 4**

```
forward(2)
right()
forward(3)
left()
forward(1)
```

Each line is one card, in the same order as the walk.

4 **Line 2: `left()`**

Line 2 turns right, but the robot should turn left.

5 **False**

Everyone makes bugs. Good programmers are good at finding and fixing them.